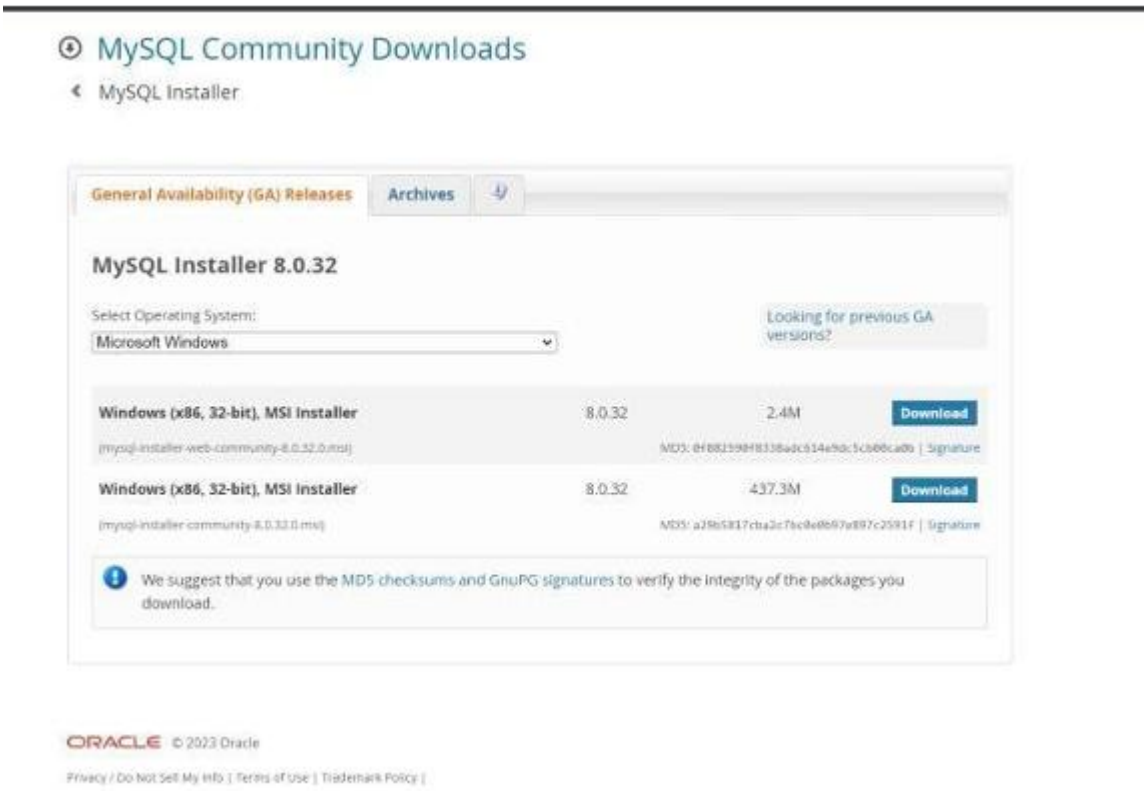
Practical 04: HIVE INSTALLATION

1. Download and install Java Development Kit (JDK) version 8

   https://adoptium.net/marketplace/?version=8

2. Download and install MySQL database server and MySQL Workbench. https://dev.mysql.com/downloads/installer/



3. The latest stable version of Apache Hive from the official website.

   https://archive.apache.org/dist/hive/hive-3.1.3/

   Install bin.tar.gz

Extract the downloaded file to a C directory.

Index of /hive/hive-3.1.3

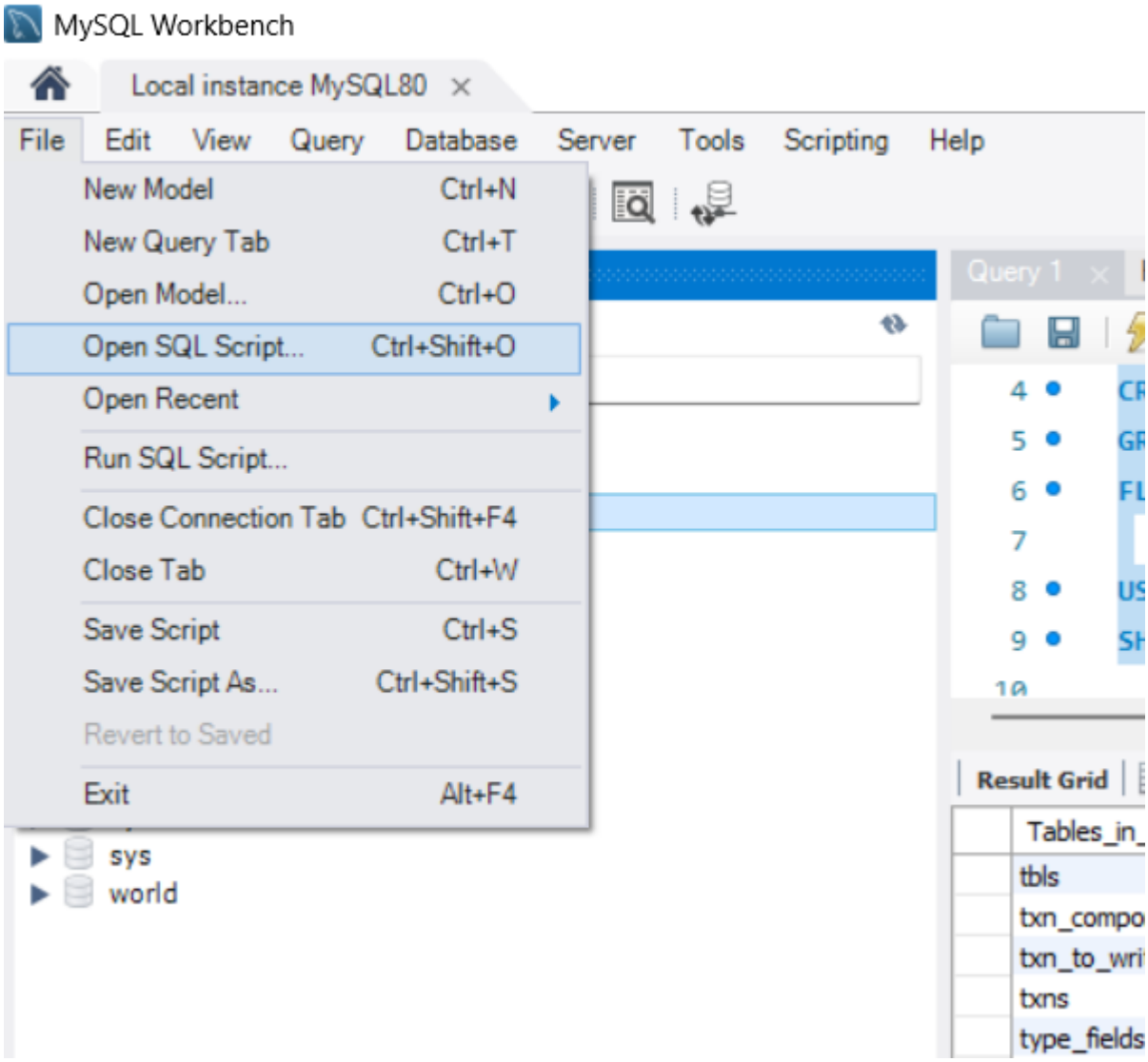
Name	Last modified	Size	Description
Parent Directory		-	
apache-hive-3.1.3-bin.tar.gz	2022-04-08 17:42	312M	
apache-hive-3.1.3-bin.tar.gz.asc	2022-04-08 17:42	488	
apache-hive-3.1.3-bin.tar.gz.sha256	2022-04-08 17:42	95	
apache-hive-3.1.3-src.tar.gz	2022-04-08 17:42	25M	
apache-hive-3.1.3-src.tar.gz.asc	2022-04-08 17:42	488	
apache-hive-3.1.3-src.tar.gz.sha256	2022-04-08 17:42	95	

4. . Install bin folder from https://github.com/HadiFadl/Hive-cmd and replace the bin in the hive folder.
5. Set the HADOOP_HOME and HIVE_HOME environment variables to the path where you extracted the downloaded files. Add till binto the PATH environment variable.
6. Open MySQL Workbench and create a new schema for Hive metastore database. Note down the schema name, username, and password.

Open workbench

```
CREATE DATABASE hive_metastore;

DROP USER IF EXISTS 'hiveuser'@'localhost';

FLUSH PRIVILEGES;

CREATE USER 'hiveuser'@'localhost' IDENTIFIED BY

'hivepassword';

GRANT ALL PRIVILEGES ON hive_metastore.* TO

'hiveuser'@'localhost';

FLUSH PRIVILEGES;


USE hive_metastore;
```

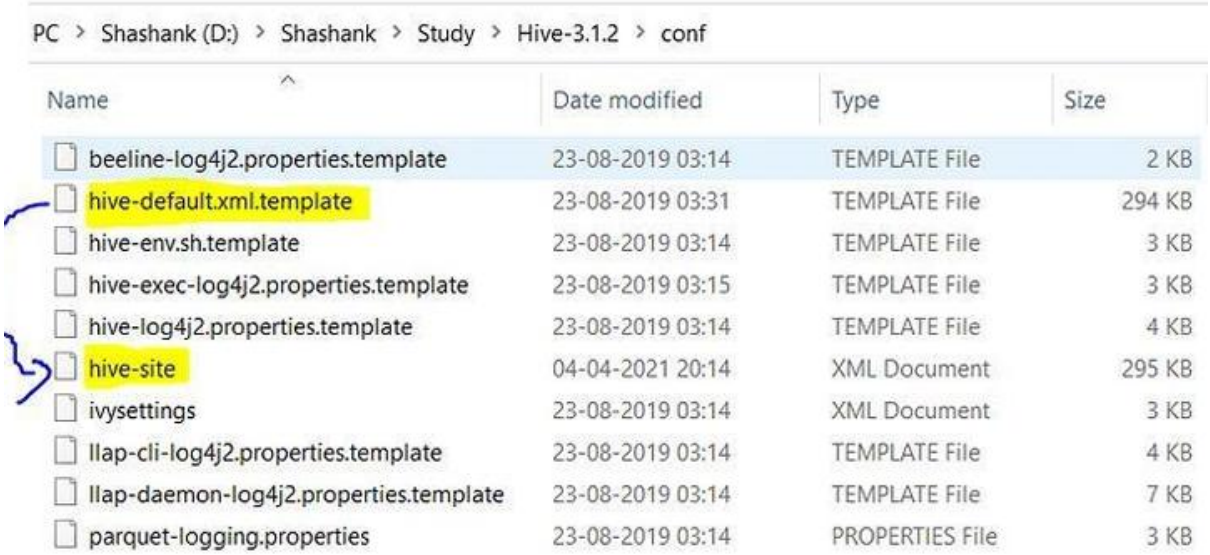
click the schema- go to file-select open sql script-open the hive-schema-3.1.0 in the workbench and execute



SHOW TABLES;

7. In the hive-site.xml file located in hive folder, update the following properties: Creating File Hive-site.xml Now we need to create the Hive-site.xml file in hive for configuring it :- (We can find these files in Hive -> conf -> hive-default.xml.template) We need to copy

the hive-default.xml.template file and paste it in the same location and rename it to hivesite.xml. This will act as our main Config file for Hive.



PC > Shashank (D:) > Shashank > Study > Hive-3.1.2 > conf

Name	Date modified	Type	Size
beeline-log4j2.properties.template	23-08-2019 03:14	TEMPLATE File	2 KB
hive-default.xml.template	23-08-2019 03:31	TEMPLATE File	294 KB
hive-env.sh.template	23-08-2019 03:14	TEMPLATE File	3 KB
hive-exec-log4j2.properties.template	23-08-2019 03:15	TEMPLATE File	3 KB
hive-log4j2.properties.template	23-08-2019 03:14	TEMPLATE File	4 KB
hive-site	04-04-2021 20:14	XML Document	295 KB
ivysettings	23-08-2019 03:14	XML Document	3 KB
llap-cli-log4j2.properties.template	23-08-2019 03:14	TEMPLATE File	4 KB
llap-daemon-log4j2.properties.template	23-08-2019 03:14	TEMPLATE File	7 KB
parquet-logging.properties	23-08-2019 03:14	PROPERTIES File	3 KB

8. Editing the hive-site.xml Remove &#8 from the hive-site

And add this in the file

<property>

<name>hive.querylog.location</name>

<value>C:/apache-hive-3.1.3-bin/bin</value>

<description>Location of Hive run time structured log file.</description>

```
  </property>

<property>
    <name>hive.exec.local.scratchdir</name>
    <value>C:/apache-hive-3.1.3-bin/bin/iotmp</value>
    <description>Local scratch space for Hive jobs.</description>
  </property>

<property>
    <name>hive.downloaded.resources.dir</name>
    <value>C:/apache-hive-3.1.3-bin/bin/iotmp</value>
    <description>Temporary local directory for added resources in the
remote file system.</description>
  </property>

<property>
    <name>hive.metastore.uris</name>
    <value>thrift://192.168.0.101:9083</value>
      </property>
```

```
<property>
    <name>javax.jdo.option.ConnectionUserName</name>
    <value>hiveuser</value>
    <description>Username to use against metastore
database.</description>
  </property>

<property>
  <name>javax.jdo.option.ConnectionURL</name>
  <value>jdbc:mysql://localhost:3306/hive_metastore</value>
  <description>JDBC connection string for Hive
metastore</description>
</property>

<property>
  <name>hive.metastore.warehouse.dir</name>
  <value>hdfs://localhost:9000/user/hive/warehouse</value>
  <description>location of default database for the
warehouse</description>
</property>
```

```
<property>
  <name>javax.jdo.option.ConnectionPassword</name>
  <value>hivepassword</value>
  <description>Password to use against metastore
database.</description>
</property>

<property>
  <name>datanucleus.schema.autoCreateSchema</name>
  <value>true</value>
</property>

<property>
  <name>datanucleus.schema.autoCreateTables</name>
  <value>True</value>
</property>

<property>
  <name>datanucleus.schema.validateTables</name>
```

```
  <value>true</value>
  <description>Validates existing schema against code.turn this on if
you want to verify existing schema</description>
</property>

<property>
  <name>javax.jdo.option.ConnectionDriverName</name>
  <value>com.mysql.cj.jdbc.Driver</value>
  <description>Driver class name for a JDBC
metastore.</description>
</property>

</configuration>
```

9. Install the .jar file from
   https://mvnrepository.com/artifact/com.mysql/mysql-connector-j/8.0.32
   and store it in the lib folder.

10. Run Hadoop nodes

11 . start cmd,

Run the command

hive --service metastore

```
PS C:\Users\sagar> hive --service metastore
"Starting Hive Metastore Server"
'C:\Program' is not recognized as an internal or external command,
operable program or batch file.
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hadoop-3.3.4/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/or
SLF4J: Found binding in [jar:file:/C:/HIVE-3.1.3/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticL
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
2023-03-17 13:03:44,755 INFO conf.MetastoreConf: Found configuration file file:/C:/HIVE-3.1.3/conf/hive-
2023-03-17 13:03:44,818 INFO conf.MetastoreConf: Unable to find config file hivemetastore-site.xml
2023-03-17 13:03:44,818 INFO conf.MetastoreConf: Found configuration file null
2023-03-17 13:03:44,819 INFO conf.MetastoreConf: Unable to find config file metastore-site.xml
2023-03-17 13:03:44,819 INFO conf.MetastoreConf: Found configuration file null
2023-03-17 13:03:44,983 WARN metastore.HiveMetaStore: Unable to initialize logging using metastore-log4j
2023-03-17 13:03:44,994 INFO metastore.HiveMetaStore: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting HiveMetaStore
STARTUP_MSG:   host = sagar/172.27.32.1
STARTUP_MSG:   args = []
STARTUP_MSG:   version = 3.1.3
STARTUP_MSG:   classpath = C:\hadoop-3.3.4\etc\hadoop;C:\hadoop-3.3.4\share\hadoop\common;C:\hadoop-3.3.
t-2.4.7.jar;C:\hadoop-3.3.4\share\hadoop\common\lib\animal-sniffer-annotations-1.17.jar;C:\hadoop-3.3.4\
```

12. open new cmd and run

"hive --service hiveserver2"

```
PS C:\Users\sagar> hive --service hiveserver2
'C:\Program' is not recognized as an internal or external command,
operable program or batch file.
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hadoop-3.3.4/share/hadoop/com
SLF4J: Found binding in [jar:file:/C:/HIVE-3.1.3/lib/log4j-slf4j-im
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFact
2023-03-17 13:03:41,108 INFO conf.HiveConf: Found configuration fil
2023-03-17 13:03:42,915 INFO server.HiveServer2: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting HiveServer2
STARTUP_MSG:   host = sagar/172.27.32.1
STARTUP_MSG:   args = []
STARTUP_MSG:   version = 3.1.3
STARTUP_MSG:   classpath = C:\hadoop-3.3.4\etc\hadoop;C:\hadoop-3.3
t-2.4.7.jar;C:\hadoop-3.3.4\share\hadoop\common\lib\animal-sniffer-
\hadoop-3.3.4\share\hadoop\common\lib\audience-annotations-0.5.0.ja
e\hadoop\common\lib\checker-qual-2.5.2.jar;C:\hadoop-3.3.4\share\ha
on\lib\commons-cli-1.2.jar;C:\hadoop-3.3.4\share\hadoop\common\lib\
tions-3.2.2.jar;C:\hadoop-3.3.4\share\hadoop\common\lib\commons-com
-2.1.1.jar;C:\hadoop-3.3.4\share\hadoop\common\lib\commons-daemon-1
```

13.open new cmd and run

hive

```
PS C:\Users\sagar> hive
'C:\Program' is not recognized as an internal or external command,
operable program or batch file.
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hadoop-3.3.4/share/hadoop/common/lib/slf4j-
SLF4J: Found binding in [jar:file:/C:/HIVE-3.1.3/lib/log4j-slf4j-impl-2.17.1.jar!
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
2023-03-17 13:03:50,490 INFO conf.HiveConf: Found configuration file file:/C:/HIV
Hive Session ID = 67e74f26-bddc-48f9-8de7-b11fad7cfce2
2023-03-17 13:03:52,259 INFO SessionState: Hive Session ID = 67e74f26-bddc-48f9-8

Logging initialized using configuration in jar:file:/C:/HIVE-3.1.3/lib/hive-commo
2023-03-17 13:03:52,524 INFO SessionState:
```

```
2026-03-15 19:52:52,388 INFO conf.HiveConf: Using
n for log id: 78ad6081-a80a-4b37-afd5-7557d56a5ef
2026-03-15 19:52:52,388 INFO session.SessionState
 main
hive>
```

Hive commands

Whenever you run command search for OK as there might be many lines showing in the output

1. CREATE (Create Database & Table)

CREATE DATABASE user_db;

USE user_db;

```
CREATE DATABASE user_db;

USE user_db;


CREATE TABLE users (

    id INT,

    name STRING,

    age INT,

    city STRING

)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ',';


show tables;
```

2. READ (Insert & Select)

```
-- Insert specific rows
```

```
INSERT INTO TABLE users VALUES (1, 'Divya', 25, 'Mumbai');

INSERT INTO TABLE users VALUES (2, 'Amit', 30, 'Delhi');


-- Read all data

SELECT * FROM users;


-- Read with a filter (WHERE clause)

SELECT name, city FROM users WHERE age > 25;
```

3. UPDATE (Overwriting Data)

Traditional UPDATE commands are slow in Hive. Instead, we usually overwrite a table or partition with new data.

```
-- This replaces the entire content of the table with a new record

INSERT OVERWRITE TABLE users VALUES (1, 'Divya', 26, 'Pune');


-- Verify the change
```

```
SELECT * FROM users;
```

4. Schema Evolution (The "Alter" Phase)

```
-- Rename a table
ALTER TABLE users RENAME TO app_users;

describe app_users;

-- Add a new column to an existing table
ALTER TABLE app_users ADD COLUMNS (email STRING,
signup_date DATE);
describe app_users;

-- Change a column name or type
ALTER TABLE app_users CHANGE COLUMN age user_age INT;
describe app_users;
```

5. DELETE (Drop Table or Database)

To remove data, you can delete specific tables or the entire database.

```
-- Delete just the table data but keep the table structure (TRUNCATE)

TRUNCATE TABLE app_users;


SELECT * FROM app_users;


-- Delete the entire table (Structure + Data)

DROP TABLE app_users;


show tables;


-- Delete the database

DROP DATABASE user_db CASCADE;


show databases;
```